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(57) Abstract:

ABSTRACT AUTONOMOUS RADAR-GUIDED SURVIVOR DETECTION AND NAVIGATION SYSTEM A system (10) for autonomous survivor detection and navigation comprises a mobile robot chassis (100), a radar sensor (110) configured to detect presence and micro-motion signatures, ultrasonic sensors (120) configured to provide distance measurements for obstacle detection, and an inertial measurement unit (130) configured to track heading data for dead-reckoning navigation. A processing unit (140) is coupled to the radar sensor (110), ultrasonic sensors (120), and inertial measurement unit (130), and maintains a grid map (200) comprising cells each having an associated confidence score. The processing unit (140) updates the confidence score based on radar detections, wherein the confidence score increases upon radar detection and decays over time absent further radar detections, and controls navigation based on the confidence scores and distance measurements. A drive mechanism (150) moves the mobile robot chassis (100) based on navigation commands from the processing unit (140).

FORM-2

**THE PATENTS ACT, 1970
(39 of 1970)**

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THE PATENTS RULES, 2003

**COMPLETE
SPECIFICATION**

10

(See section 10 and rule 13)

TITLE OF INVENTION

**AUTONOMOUS RADAR-GUIDED SURVIVOR DETECTION AND
NAVIGATION SYSTEM**

15

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**THE FOLLOWING SPECIFICATION PARTICULARLY DESCRIBES
THE INVENTION AND THE MANNER IN WHICH IT IS TO BE
PERFORMED**

CROSS-REFERENCE TO RELATED APPLICATIONS AND PRIORITY

[0001] The present application does not claim priority from any patent application.

PREAMBLE

5 [0002] The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF INVENTION

[0003] The present disclosure relates to autonomous mobile robotic systems for search and rescue operations, and more particularly to a radar-guided navigation
10 and survivor detection system that employs probabilistic confidence mapping and sensor fusion for locating survivors in disaster environments.

BACKGROUND

[0004] The following text of background art is provided for purposes of understanding and is not an admission that any cited document or technique forms
15 part of the prior art under applicable law.

[0005] Natural and man-made disasters such as earthquakes, building collapses, landslides, and floods frequently result in survivors becoming trapped under debris in locations that are inaccessible to rescue teams. The areas where survivors may be trapped under rubble or debris can be challenging for rescue
20 personnel, canines, or other rescue resources to reach, as they may not be able to thoroughly navigate through rough terrain, crevices, nooks, and corners under the debris.

[0006] Conventional search and rescue methods have relied on manual search techniques, individual sensor tools, or robotic systems for detecting trapped
25 survivors. However, these approaches face multiple challenges including obstructed visibility, cluttered debris, unstable surfaces, and extreme time constraints. Traditional rescue equipment using cameras, infrared thermal sensors, or passive infrared (PIR) sensors may detect humans or animals only when they are moving and may not be able to detect survivors while they are stationary or
30 unconscious. Furthermore, visual sensors and laser-based ranging systems are

susceptible to interference from environmental conditions such as dust, smoke, fog, and low-light conditions commonly encountered in disaster scenarios.

[0007] US 2024/0077873 A1 (Pannala) discloses a radar sensor-based bio-inspired autonomous mobile robot using BLE location tracking for disaster rescue.

5 The system incorporates millimeter-wave radar sensors and Ultra Wideband (IR-UWB) radar sensors for detecting presence and performing ranging of trapped survivors, along with GNSS and BLE connectivity for location tracking. The mobile robot uses bio-inspired jointed segmented limbs driven by actuators for locomotion through challenging terrain.

10 **[0008]** IN 202511119648 A discloses an integrated multi-sensor system with Edge-AI for survivor detection and rescuer guidance in disaster environments. The system integrates ultrasonic, PIR, sound, temperature, and rain sensors combined with a servo-rotating ultrasonic radar for 360-degree scanning and a camera module for visual confirmation. A machine learning model processes sensor data to classify
15 human presence and generate directional guidance.

[0009] CN 114063628 B discloses a UAV-based disaster relief system using deep learning and radar detection. The system employs microwave radar to detect human presence through walls and debris, with a neural network model trained to identify trapped personnel. The UAV adjusts its search trajectory based on
20 detection feedback from the radar system.

[0010] CN 112731321 B discloses a mobile robot obstacle avoidance and map generation method based on MIMO cognitive radar. The system introduces a target certainty memory grid map that enables real-time obstacle avoidance using accumulated radar measurement data. The method stores and dynamically increases
25 obstacle certainty values to build confidence in obstacle presence over multiple measurement frames.

[0011] CN 117990085 A discloses an autonomous exploration map generation method based on dual millimeter-wave radar. The system addresses limitations of visual sensors and laser radar in smoke-filled emergency rescue scenarios by
30 utilizing millimeter-wave radar for autonomous exploration and scene mapping in environments where conventional sensors fail.

[0012] Existing systems may suffer from limited sensing capability due to reliance on single-sensor inputs, high rates of false positives from environmental noise and non-human heat sources, slow decision-making due to scattered data streams, lack of real-time guidance for optimal movement paths, and inadequate performance under low-visibility conditions. Many prior approaches also rely on complex simultaneous localization and mapping (SLAM) techniques, camera-based systems, or light detection and ranging (LiDAR) data that may be computationally expensive or unreliable in disaster environments characterized by dust, smoke, and structural damage.

[0013] The present invention addresses the above shortcomings of the prior art. However, the invention is entirely different from the prior art in terms of novelty and technological advancements.

OBJECT

[0014] In view of the foregoing, and without prejudice to the generality of the disclosure, it is an object of the present invention to provide. It is to be understood that the foregoing objects are illustrative and non-limiting; additional objects will be apparent from the description, examples, and claims, and need not be achieved in every embodiment.

[0015] The object of the present disclosure is to provide an autonomous mobile robotic system for survivor detection and navigation in disaster rescue environments that overcomes the limitations of existing camera-based and LiDAR-dependent approaches. The system aims to enable reliable operation in low-visibility conditions such as smoke, dust, fog, and darkness where conventional visual and laser-based sensors may fail to function effectively.

[0016] A further object of the present disclosure is to provide a radar-guided navigation and detection system that employs a probabilistic survivor confidence mapping approach, wherein radar returns are accumulated over time to build a real-time heatmap indicating likely survivor locations. By maintaining and updating confidence scores for grid cells based on repeated radar detections, the system seeks to reduce false positives and improve localization accuracy compared to single-reading detection methods.

[0017] Another object of the present disclosure is to provide an autonomous navigation system that operates without reliance on simultaneous localization and mapping (SLAM) techniques, camera data, or light detection and ranging (LiDAR) data. The system aims to achieve autonomous grid navigation using wall-following exploration strategies combined with inertial measurement unit-based dead-reckoning for heading estimation, thereby reducing computational complexity and enabling deployment on resource-constrained processing platforms.

[0018] Yet another object of the present disclosure is to provide a threshold-driven state machine architecture that governs robot behaviour based on accumulated radar confidence scores. The state machine aims to transition the robot between exploration, confirmation, and logging states in response to confidence thresholds, enabling the system to pause exploration when a potential survivor is detected, perform directional sweeps to verify the detection, and log confirmed survivor locations with associated position data.

[0019] A still further object of the present disclosure is to provide a system capable of detecting, clustering, and ranking multiple survivor locations simultaneously based on aggregate confidence scores, thereby enabling rescue teams to prioritize response efforts according to detection confidence levels.

[0020] For the avoidance of doubt, the foregoing objects encompass compositions and manufacturing processes suitable for adoption in jurisdictions where claims to methods of treatment of animals may be restricted; corresponding “use” or “product” claim formats are contemplated without limitation.

SUMMARY

[0021] The following summary is intended to introduce aspects of the present invention in a simplified form and does not identify all the features or the scope of the claimed invention. The summary is provided to assist in understanding the invention and is to be read in conjunction with the detailed description and accompanying drawings, claims, and abstract. One or more embodiments of the present disclosure are set forth below solely for illustrative purposes and should not be construed as limiting.

[0022] According to an aspect of the present disclosure, a system for autonomous survivor detection and navigation in a disaster rescue environment is provided. The system includes a mobile robot chassis. A radar sensor is disposed on the mobile robot chassis and configured to detect presence and micro-motion signatures within a detection range. A plurality of ultrasonic sensors are distributed around the mobile robot chassis and configured to provide distance measurements for obstacle detection. An inertial measurement unit is disposed on the mobile robot chassis and configured to track heading data for dead-reckoning navigation. A processing unit is communicatively coupled to the radar sensor, the plurality of ultrasonic sensors, and the inertial measurement unit. The processing unit is configured to maintain a grid map comprising a plurality of cells, each cell having an associated confidence score. The processing unit is configured to update the confidence score of a given cell based on radar detections from the radar sensor, wherein the confidence score increases in response to a radar detection associated with the given cell and decays over time in an absence of further radar detections. The processing unit is configured to control navigation of the mobile robot chassis based on the confidence scores and the distance measurements from the plurality of ultrasonic sensors. A drive mechanism is operatively connected to the processing unit and configured to move the mobile robot chassis based on navigation commands from the processing unit.

[0023] The combination of radar-based detection with probabilistic confidence mapping provides technical advantages in disaster rescue scenarios. The radar sensor enables detection of survivors through obstacles such as debris and rubble where visual sensors would fail. The confidence score mechanism that increases upon detection and decays over time reduces false positives by requiring consistent radar returns before confirming a survivor location. The sensor fusion approach combining radar, ultrasonic, and inertial measurement data enables robust navigation and detection in low-visibility conditions characterized by dust, smoke, and darkness. The grid map structure provides a spatial representation that allows the system to track and localize potential survivors across the search environment.

[0024] According to other aspects of the present disclosure, the processing unit may be further configured to operate a state machine comprising an exploration state in which the mobile robot chassis navigates the environment using wall-following and dead-reckoning, a confirmation state in which the mobile robot chassis performs a directional sweep to gather additional radar readings when the confidence score of a cell exceeds a first threshold, and a logging state in which the processing unit records a confirmed survivor location when the confidence score remains above a second threshold during the directional sweep.

[0025] The state machine architecture provides technical advantages by enabling structured decision-making based on accumulated confidence scores. The exploration state allows systematic coverage of the disaster environment. The confirmation state reduces false positives by requiring additional verification before logging a survivor location. The logging state ensures that confirmed survivor locations are recorded with associated position data for rescue team coordination. The threshold-driven transitions enable the system to balance exploration efficiency with detection accuracy.

[0026] According to other aspects of the present disclosure, the processing unit may be configured to transition from the exploration state to the confirmation state when the confidence score of a cell exceeds the first threshold, and to transition from the confirmation state to the logging state when a confirmation criterion is satisfied.

[0027] The threshold-based state transitions provide technical advantages by automating the decision process for survivor detection verification. This approach enables the system to autonomously determine when sufficient evidence has accumulated to warrant further investigation and when a detection has been sufficiently confirmed to be logged as a survivor location.

[0028] According to other aspects of the present disclosure, the processing unit may be configured to update the confidence score of the given cell according to a formula that increases the confidence score by an increment value when a radar detection is associated with the given cell, and decreases the confidence score by a

decay value multiplied by an elapsed time interval when no radar detection is associated with the given cell.

[0029] The formula-based confidence score update mechanism provides technical advantages by enabling quantitative tracking of detection confidence over
5 time. The increment-based increase upon detection allows confidence to build with repeated radar returns. The time-based decay ensures that stale detections do not persist indefinitely, thereby maintaining an accurate representation of current survivor likelihood across the grid map.

[0030] According to other aspects of the present disclosure, the plurality of
10 ultrasonic sensors may comprise a front ultrasonic sensor, a left ultrasonic sensor, and a right ultrasonic sensor, and the processing unit may be configured to implement a wall-following exploration strategy using the distance measurements from the plurality of ultrasonic sensors.

[0031] The distributed ultrasonic sensor arrangement provides technical
15 advantages by enabling obstacle detection across multiple directions simultaneously. The front, left, and right sensor configuration supports wall-following navigation strategies that enable systematic exploration of enclosed spaces and corridors commonly encountered in disaster environments. This arrangement allows the system to maintain safe distances from obstacles while
20 navigating through debris fields.

[0032] According to other aspects of the present disclosure, the processing unit may be configured to control navigation of the mobile robot chassis without reliance on simultaneous localization and mapping, camera data, or light detection and ranging data.

[0033] The navigation approach that operates without reliance on simultaneous
25 localization and mapping, camera data, or light detection and ranging data provides technical advantages in disaster rescue scenarios. This approach reduces computational complexity, enabling deployment on resource-constrained processing platforms. The independence from visual and laser-based sensors
30 enables reliable operation in environments characterized by dust, smoke, fog, and darkness where such sensors may fail.

[0034] According to other aspects of the present disclosure, the radar sensor may comprise a millimeter-wave radar module configured to detect micro-motion signatures indicative of breathing patterns.

[0035] The millimeter-wave radar module configured to detect micro-motion signatures indicative of breathing patterns provides technical advantages for survivor detection. This capability enables detection of stationary or unconscious survivors who may not exhibit gross movement. The ability to detect breathing patterns through obstacles such as debris and rubble extends the detection capability beyond line-of-sight scenarios.

[0036] According to other aspects of the present disclosure, the processing unit may be configured to identify candidate survivor locations by grouping neighbouring cells of the grid map having confidence scores that exceed a detection threshold, and to rank the candidate survivor locations based on aggregate confidence scores.

[0037] The grouping and ranking mechanism provides technical advantages by enabling identification of multiple potential survivor locations simultaneously. The clustering of neighbouring high-confidence cells improves localization accuracy by aggregating spatially related detections. The ranking based on aggregate confidence scores enables rescue teams to prioritize response efforts according to detection confidence levels.

[0038] According to other aspects of the present disclosure, the processing unit may be configured to execute a periodic control loop at a fixed rate, wherein each iteration of the control loop comprises receiving radar return data, updating confidence scores for cells of the grid map, receiving distance measurements from the plurality of ultrasonic sensors, and generating navigation commands.

[0039] The periodic control loop executed at a fixed rate provides technical advantages by ensuring consistent and predictable system behaviour. The structured iteration comprising radar data reception, confidence score updates, ultrasonic distance measurements, and navigation command generation enables real-time responsiveness to changing environmental conditions. The fixed-rate execution facilitates coordination between sensor data acquisition and navigation control.

[0040] According to other aspects of the present disclosure, the directional sweep may comprise the radar sensor sampling across a plurality of angular directions relative to a current heading of the mobile robot chassis as determined by the inertial measurement unit.

5 [0041] The directional sweep comprising radar sampling across a plurality of angular directions provides technical advantages by enabling verification of potential survivor detections from multiple perspectives. The correlation of radar sampling directions with heading information from the inertial measurement unit enables accurate spatial mapping of radar returns. This multi-angle sampling
10 approach improves localization accuracy and reduces the likelihood of false positives arising from single-direction measurements.

[0042] The foregoing general description of the illustrative embodiments and the following detailed description thereof are merely exemplary aspects of the teachings of this disclosure and are not restrictive.

15 BRIEF DESCRIPTION OF FIGURES

[0043] The drawings appended herein illustrate exemplary embodiments of the present invention and are provided to enhance understanding of the inventive features disclosed. These figures, when read in conjunction with the detailed description, depict the operational flow, component architecture, and other
20 mechanisms.

[0044] The illustrations are schematic in nature and serve solely to clarify the principles and functionality of the present disclosure. While particular constructions and configurations are depicted for explanatory purposes, it will be understood that the invention is not limited to the precise implementations shown in the figures and
25 may encompass variations without departing from the scope of protection.

[0045] FIG. 1 illustrates a block diagram of a system for autonomous radar-guided survivor detection and navigation, according to aspects of the present disclosure.

[0046] FIG. 2 depicts a grid map representing a probabilistic survivor confidence map for the system of FIG. 1, according to an embodiment.
30

[0047] FIG. 3 illustrates a flowchart for a periodic control loop for controlling navigation of a mobile robot chassis, according to aspects of the present disclosure.

[0048] FIG. 4 illustrates a flowchart depicting a state machine for survivor detection and confirmation, according to an embodiment.

5 [0049] FIG. 5 depicts a plan view of a radar sampling configuration for a mobile robot operating within a grid map, according to aspects of the present disclosure.

[0050] FIG. 6 depicts a grid map illustrating candidate survivor locations identified through grouping of neighbouring cells, according to an embodiment.

10 [0051] FIG. 7 illustrates an exploded view of a mobile robot chassis showing an arrangement of components, according to aspects of the present disclosure.

[0052] Furthermore, in terms of the construction of the system, one or more components of the device may have been represented in the figures by conventional symbols, and the figures may show only those specific details that are pertinent to understanding the embodiments of the present invention so as not to obscure the figures with details that will be readily apparent to those of ordinary skill in the art having benefit of the description herein.

DETAILED DESCRIPTION

[0053] The following detailed description is intended to provide an in-depth understanding of the invention and its various components, configurations, and operational aspects. While specific embodiments are described with reference to the accompanying drawings, the invention is not limited to the particular forms disclosed herein. The scope of the invention shall be interpreted broadly and in accordance with the claims appended hereto.

25 [0054] The terms “comprises,” “comprising,” “includes,” “including,” and similar expressions used in this description are intended to be open-ended and should be interpreted to include, but not be limited to, the mentioned elements. Unless explicitly stated otherwise, singular terms may include their plural equivalents and vice versa, depending on the context in which they appear.

30 [0055] Reference is now made to the accompanying figures, which illustrate example embodiments of the present invention. It should be appreciated that the

figures are intended for illustration and explanatory purposes only and do not limit the scope of the invention. Similar reference numerals have been used to denote functionally similar components throughout the figures.

5 [0056] The embodiments described herein are presented for illustrative purposes and are subject to variations, modifications, and adaptations by those skilled in the art. Any equivalent implementations that perform substantially the same function in substantially the same manner are intended to fall within the scope of this disclosure and the claims appended hereto.

10 [0057] The use of the term “exemplary” in the description shall be understood to mean “serving as an example or illustration” and shall not be construed as limiting the invention to the preferred embodiments disclosed. Various other embodiments may be developed without departing from the spirit or essential characteristics of the invention. The technical principles and advantages outlined in the foregoing description should be read in conjunction with the figures and claims.

15 It is expressly understood that while the detailed description sets out specific structural and operational aspects, the scope of protection is governed by the claims and includes all legal equivalents of the claimed subject matter. Embodiments of the disclosure are described in the following paragraphs with reference to Figures. In Figures, the same elements or elements that have the same functions are indicated

20 by the same reference signs.

[0058] The present invention will now be described in detail with reference to the accompanying drawings illustrating exemplary embodiments.

25 [0059] FIG. 1 illustrates a block diagram of a system 10 for autonomous survivor detection and navigation in a disaster rescue environment according to various embodiments.

[0060] The system 10 includes a mobile robot chassis 100 that serves as a structural platform for housing and supporting the various components of the system 10. The mobile robot chassis 100 provides a base structure upon which sensors, processing electronics, and locomotion mechanisms are mounted.

30 [0061] A radar sensor 110 is disposed on the mobile robot chassis 100 and configured to detect presence and micro-motion signatures within a detection range.

The radar sensor 110 may comprise a millimeter-wave radar module operating at frequencies such as 24 GHz. In some cases, the radar sensor 110 may be modeled after an LD2410C radar module with detection characteristics suitable for presence and micro-motion detection through walls and rubble. The radar sensor 110 may have a detection probability of approximately 0.85 and a false-positive rate of approximately 0.05 for realistic sensing performance. In some cases, the radar sensor 110 may detect moving targets at approximately 0.30 m range and static targets at approximately 0.72 m range. The radar sensor 110 may comprise a frequency-modulated continuous wave (FMCW) radar or other suitable radar technologies. The millimeter-wave radar module is configured to detect micro-motion signatures indicative of breathing patterns, enabling detection of stationary or unconscious survivors who may not exhibit gross movement. The radar sensor 110 is capable of detecting presence through walls and rubble using non-line-of-sight detection capabilities.

15 **[0062]** A plurality of ultrasonic sensors 120 are distributed around the mobile robot chassis 100 and configured to provide distance measurements for obstacle detection. The plurality of ultrasonic sensors 120 may comprise a front ultrasonic sensor 120, a left ultrasonic sensor 120, and a right ultrasonic sensor 120. In some cases, the ultrasonic sensors 120 may be modeled after HC-SR04 sensors with a detection range of approximately 2-400 cm. The ultrasonic sensors 120 may comprise piezoelectric transducers, time-of-flight sensors, or other proximity sensing mechanisms suitable for distance measurement in disaster environments.

20 **[0063]** An inertial measurement unit 130 is disposed on the mobile robot chassis 100 and configured to track heading data for dead-reckoning navigation. The inertial measurement unit 130 enables the system 10 to maintain positional awareness without reliance on external mapping systems. In some cases, the inertial measurement unit 130 may comprise microelectromechanical systems (MEMS) gyroscopes, accelerometers, magnetometers, or sensor fusion configurations that combine data from multiple inertial sensors.

25 **[0064]** A processing unit 140 is communicatively coupled to the radar sensor 110, the plurality of ultrasonic sensors 120, and the inertial measurement unit 130.

30

The processing unit 140 receives sensor data from the radar sensor 110, the ultrasonic sensors 120, and the inertial measurement unit 130, and executes navigation algorithms and survivor detection logic. In some cases, the processing unit 140 may comprise a dual-core processor operating at approximately 240 MHz with under 200 KB RAM, modeled after ESP32 microcontroller constraints. The processing unit 140 may comprise microcontrollers, embedded processors, or system-on-chip devices suitable for resource-constrained applications. The processing unit 140 operates without cloud connectivity dependency, executing all processing locally on the mobile robot chassis 100.

10 **[0065]** The processing unit 140 is configured to control navigation of the mobile robot chassis 100 without reliance on simultaneous localization and mapping (SLAM), camera data, or light detection and ranging (LiDAR) data. The system 10 detects survivors by identifying breathing patterns through micro-motion signatures captured by the radar sensor 110.

15 **[0066]** A drive mechanism 150 is operatively connected to the processing unit 140 and configured to move the mobile robot chassis 100 based on navigation commands from the processing unit 140. In some cases, the drive mechanism 150 may comprise an L298N-driven differential drive system for motor control. The drive mechanism 150 may comprise differential drive systems, omnidirectional
20 wheels, tracked drives, or other suitable locomotion mechanisms. The drive mechanism 150 includes wheels and associated motor components positioned at a lower portion of the mobile robot chassis 100 for enabling movement through the disaster rescue environment.

25 **[0067]** FIG. 2 illustrates a grid map 200 representing a probabilistic survivor confidence map according to various embodiments.

30 **[0068]** The processing unit 140 maintains the grid map 200 comprising a plurality of cells, each cell having an associated confidence score. The grid map 200 may be organized as a two-dimensional array of cells arranged in rows and columns. The mobile robot chassis 100 is positioned within the grid map 200, and a radar detection range boundary surrounding the mobile robot chassis 100 represents the area within which the radar sensor 110 may sense potential survivors.

The grid map 200 includes obstacle cells representing walls or debris, and free-space cells representing navigable areas.

[0069] The processing unit 140 updates the confidence score of a given cell based on radar detections from the radar sensor 110. The confidence score increases
5 in response to a radar detection associated with the given cell. In some cases, the confidence score of the given cell may be updated according to the formula $C = \min(1.0, C + 0.12 \times r)$, where C represents the confidence score and r represents a reliability factor associated with the radar detection. The reliability factor r may account for signal strength, detection consistency, or other parameters affecting the
10 quality of the radar return.

[0070] The confidence score decays over time in an absence of further radar detections. The processing unit 140 may decrease the confidence score by a decay value multiplied by an elapsed time interval when no radar detection is associated with the given cell. In some cases, the decay may be computed according to the
15 formula $C = \max(0, C - \beta \times \Delta t)$, where β represents the decay value and Δt represents the elapsed time interval since the previous update. The decay mechanism ensures that stale detections do not persist indefinitely and that the grid map 200 reflects current conditions within the disaster rescue environment.

[0071] The confidence score scale ranges from 0.0 to 1.0, with corresponding
20 shading patterns indicating the likelihood of survivor presence. Low confidence scores approaching 0.0 may be represented by white or light shading, medium confidence scores approaching 0.5 may be represented by intermediate gray shading, and high confidence scores approaching 1.0 may be represented by dark or black shading. Cells within candidate survivor clusters that meet or exceed a
25 threshold value may be identified as potential survivor locations.

[0072] The processing unit 140 controls navigation of the mobile robot chassis 100 based on the confidence scores and the distance measurements from the plurality of ultrasonic sensors 120. The grid map 200 provides a spatial representation that allows the system 10 to track and localize potential survivors
30 across the search environment. The processing unit 140 may display a real-time probability heatmap visualization, where locations with highest survivor

probability based on breathing pattern detection may be indicated by visual markers such as pink dots or other distinguishing indicators. The real-time probability heatmap visualization enables operators or rescue personnel to identify areas of interest within the disaster rescue environment based on accumulated radar confidence data.

[0073] FIG. 3 illustrates a flowchart for a periodic control loop executed at a fixed rate according to various embodiments.

[0074] The processing unit 140 is configured to execute a periodic control loop at a fixed rate. In some cases, the processing unit 140 may execute the control loop at a rate of 10 Hz. Each iteration of the control loop comprises receiving radar return data from the radar sensor 110, updating confidence scores for cells of the grid map 200, receiving distance measurements from the plurality of ultrasonic sensors 120, and generating navigation commands.

[0075] During each iteration, the processing unit 140 receives radar return data from the radar sensor 110 for a plurality of cells of the grid map 200. The processing unit 140 performs range-Doppler processing and cell mapping to associate radar returns with corresponding cells of the grid map 200. The processing unit 140 determines whether a radar detection is associated with a given cell. When a radar detection is associated with the given cell, the processing unit 140 increases the confidence score for the cell. When no radar detection is associated with the given cell, the processing unit 140 decays the confidence score for the cell over time in the absence of further radar detections, as described above.

[0076] Following the confidence score update, the processing unit 140 receives distance measurements from the plurality of ultrasonic sensors 120 positioned at front, left, and right locations on the mobile robot chassis 100. The processing unit 140 generates navigation commands based on the confidence scores and the distance measurements from the plurality of ultrasonic sensors 120.

[0077] The processing unit 140 is configured to implement a wall-following exploration strategy using the distance measurements from the plurality of ultrasonic sensors 120. In some cases, the processing unit 140 may implement a right-hand wall-following rule for autonomous navigation. The right-hand wall-

following rule directs the mobile robot chassis 100 to maintain proximity to a wall or obstacle on the right side while navigating through the disaster rescue environment. The processing unit 140 uses heading estimation from the inertial measurement unit 130 to maintain orientation during wall-following navigation.

5 The combination of distance measurements from the front, left, and right ultrasonic sensors 120 with heading data from the inertial measurement unit 130 enables the mobile robot chassis 100 to navigate systematically through enclosed spaces and corridors.

[0078] FIG. 4 illustrates a flowchart depicting a state machine for survivor
10 detection and confirmation according to various embodiments.

[0079] The processing unit 140 is configured to operate a state machine comprising an exploration state, a confirmation state, and a logging state. The state machine governs the behavior of the mobile robot chassis 100 based on accumulated radar confidence scores.

15 [0080] In the exploration state, the mobile robot chassis 100 navigates the environment using wall-following and dead-reckoning. During the exploration state, the radar sensor 110 continuously monitors for potential survivor signatures while the mobile robot chassis 100 traverses the disaster rescue environment. The processing unit 140 accumulates radar detection data and updates confidence scores
20 for grid cells within the detection range of the radar sensor 110.

[0081] The processing unit 140 is configured to transition from the exploration state to the confirmation state when the confidence score of a cell exceeds a first threshold. In some cases, the first threshold may be set at a confidence score of 0.72. When the confidence score of a cell exceeds the first threshold, the processing
25 unit 140 transitions the state machine to the confirmation state.

[0082] In the confirmation state, the mobile robot chassis 100 performs a directional sweep to gather additional radar readings. The mobile robot chassis 100 pauses exploration movement and performs the directional sweep to verify the initial detection and refine the estimated location of the potential survivor. The
30 directional sweep allows the system 10 to gather radar readings from multiple angles relative to the detected cell.

- [0083] The processing unit 140 is configured to transition from the confirmation state to the logging state when a confirmation criterion is satisfied. The confirmation criterion may comprise the confidence score remaining above a second threshold during the directional sweep. When the confirmation criterion is not satisfied during the directional sweep, the processing unit 140 returns the state machine to the exploration state to continue searching the environment.
- [0084] In the logging state, the processing unit 140 records a confirmed survivor location when the confidence score remains above the second threshold during the directional sweep. The processing unit 140 records the confirmed survivor location along with associated position data. Once the survivor location is logged, the processing unit 140 transitions the state machine back to the exploration state to continue searching for additional survivors in unexplored areas of the environment.
- [0085] The threshold-driven state transitions enable the system 10 to balance exploration efficiency with detection accuracy by requiring confirmation before logging survivor locations.
- [0086] FIG. 5 illustrates a plan view of a radar sampling configuration for the mobile robot chassis 100 performing a directional sweep within the grid map 200 according to various embodiments.
- [0087] The directional sweep comprises the radar sensor 110 sampling across a plurality of angular directions relative to a current heading of the mobile robot chassis 100 as determined by the inertial measurement unit 130. The radar sampling directions extend radially from the mobile robot chassis 100 at angular intervals. In some cases, the angular intervals may include negative ninety degrees, negative sixty degrees, negative forty-five degrees, negative thirty degrees, negative fifteen degrees, zero degrees, fifteen degrees, thirty degrees, forty-five degrees, sixty degrees, and ninety degrees relative to the current heading. An additional sampling direction at one hundred eighty degrees may be positioned directly behind the mobile robot chassis 100.
- [0088] During the confirmation state, the directional sweep may comprise a full 360-degree rotation to gather radar readings from all angular directions. The

processing unit 140 correlates radar returns from each sampling direction with heading information from the inertial measurement unit 130 to enable accurate spatial mapping of radar returns to cells of the grid map 200. The multi-angle sampling approach enables the system 10 to verify potential survivor detections from multiple perspectives and reduces the likelihood of false positives arising from single-direction measurements.

[0089] When the mobile robot chassis 100 reaches a high probability location within the grid map 200, the processing unit 140 may perform triangulation to confirm the exact location of a detected survivor. The triangulation process combines radar readings from the plurality of angular directions with position data from the inertial measurement unit 130 to refine the estimated survivor location within the grid map 200.

[0090] FIG. 6 illustrates the grid map 200 with candidate survivor locations identified through grouping of neighbouring cells according to various embodiments.

[0091] The processing unit 140 is configured to identify candidate survivor locations by grouping neighbouring cells of the grid map 200 having confidence scores that exceed a detection threshold. The processing unit 140 analyzes the grid map 200 to identify clusters of adjacent cells where confidence scores indicate potential survivor presence. In some cases, the detection threshold may be set at a confidence score value that distinguishes between background noise and sustained radar detections indicative of survivor presence.

[0092] The processing unit 140 is configured to rank the candidate survivor locations based on aggregate confidence scores. The aggregate confidence score for each candidate survivor location may be computed by combining the confidence scores of the grouped neighbouring cells within the cluster. In some cases, the aggregate confidence score may comprise an average, a sum, or a weighted combination of the individual cell confidence scores within the cluster.

[0093] The grid map 200 may display a plurality of candidate survivor locations, each represented by a cluster of cells with varying confidence scores. In some cases, the confidence scores may be represented by shading patterns, where

cells with confidence scores in a range of 0.75 to 1.00 may be represented by dark shading indicating high confidence, cells with confidence scores in a range of 0.50 to 0.75 may be represented by diagonal line shading, cells with confidence scores in a range of 0.25 to 0.50 may be represented by cross-hatched shading, and cells with confidence scores in a range of 0.00 to 0.25 may be represented by dotted shading indicating low confidence.

[0094] The processing unit 140 may generate a ranking table listing the candidate survivor locations ordered by aggregate confidence scores. In some cases, the ranking table may include a rank identifier, a candidate identifier, and an aggregate confidence score for each candidate survivor location. The ranking enables rescue teams to prioritize response efforts according to detection confidence levels. At the end of a mission, the processing unit 140 may cluster neighbouring high-confidence cells and rank confirmed survivor locations by confidence score to provide a prioritized list of locations for rescue personnel.

[0095] FIG. 7 illustrates an exploded view of the mobile robot chassis 100 showing the arrangement and relationship of components according to various embodiments.

[0096] The mobile robot chassis 100 comprises a radar sensor 110 positioned at a top portion of the assembly. The radar sensor 110 is configured to detect presence and micro-motion signatures through obstacles such as walls and rubble. The positioning of the radar sensor 110 at the top portion of the mobile robot chassis 100 may provide an unobstructed field of view for detecting survivors in the disaster rescue environment.

[0097] An inertial measurement unit 130 is mounted adjacent to the radar sensor 110 on the mobile robot chassis 100. The inertial measurement unit 130 is configured to track heading data for dead-reckoning navigation. The proximity of the inertial measurement unit 130 to the radar sensor 110 may facilitate coordination between heading data and radar sampling directions during directional sweeps.

[0098] Three ultrasonic sensors 120 are positioned on the mobile robot chassis 100 to provide multi-directional obstacle detection. A front ultrasonic sensor 120 is

positioned at a forward location on the mobile robot chassis 100. A left ultrasonic sensor 120 is positioned on a left side of the mobile robot chassis 100. A right ultrasonic sensor 120 is positioned on a right side of the mobile robot chassis 100. The distributed arrangement of the ultrasonic sensors 120 enables obstacle
5 detection across multiple directions and supports wall-following navigation strategies.

[0099] A processing unit 140 is housed within the mobile robot chassis 100. The processing unit 140 executes control algorithms for navigation and survivor detection. In some cases, the processing unit 140 may operate within defined
10 compute and memory constraints. In some cases, the processing unit 140 may comprise a dual-core processor operating at approximately 240 MHz with under 200 KB RAM. The processing unit 140 operates without cloud connectivity dependency, executing all processing locally on the mobile robot chassis 100.

[0100] A drive mechanism 150 is positioned at a lower portion of the mobile
15 robot chassis 100. The drive mechanism 150 includes wheels and motor components for enabling movement of the mobile robot chassis 100 through the disaster rescue environment. In some cases, the drive mechanism 150 may comprise a differential drive system for motor control.

[0101] A power source is located beneath the mobile robot chassis 100 to
20 supply electrical power to the radar sensor 110, the ultrasonic sensors 120, the inertial measurement unit 130, the processing unit 140, and the drive mechanism 150. In some cases, the power source may comprise a lithium-ion battery or other rechargeable battery configurations. In some cases, the power source may comprise a lithium polymer battery, a nickel-metal hydride battery, or other portable power
25 storage devices suitable for mobile robotic applications.

[0102] The system 10 may include a GPS module configured to log GPS coordinates when a survivor location is confirmed. The GPS module may be communicatively coupled to the processing unit 140. When the processing unit 140 transitions to the logging state and records a confirmed survivor location, the GPS
30 module may provide coordinate data associated with the confirmed survivor location. In some cases, the GPS module may output NMEA data over a UART

interface. The NMEA data may include latitude, longitude, and timestamp information corresponding to the confirmed survivor location. The GPS coordinate data may be stored by the processing unit 140 along with the confidence score and grid map position data for the confirmed survivor location.

5 **[0103]** In some cases, the assembly configuration of the mobile robot chassis 100 may vary based on deployment requirements. The radar sensor 110 may be mounted at alternative positions on the mobile robot chassis 100, such as at a forward-facing location or at an elevated position using a mounting bracket. The ultrasonic sensors 120 may be positioned at alternative angular orientations to
10 provide different coverage patterns for obstacle detection. In some cases, additional ultrasonic sensors 120 may be positioned at rear locations on the mobile robot chassis 100 to provide rearward obstacle detection capability.

[0104] In some cases, the power source arrangement may vary based on mission duration requirements. Multiple battery cells may be arranged in series or
15 parallel configurations to provide increased voltage or capacity. The power source may include power management circuitry for monitoring battery state and distributing power to the various components of the mobile robot chassis 100.

[0105] It is to be appreciated that the concepts, systems, and techniques sought to be protected herein are not limited to use in the example applications described
20 herein, but rather may be useful in substantially any application where it is desired to detect and locate persons in environments with limited visibility or obstructed access. The system 10 for autonomous survivor detection and navigation is susceptible of industrial application in the emergency response and disaster management industry, where autonomous mobile robotic systems may be deployed
25 for search and rescue operations following natural disasters such as earthquakes, building collapses, landslides, and floods. The radar-guided navigation and probabilistic confidence mapping approach may find application in civil defense organizations, fire and rescue services, military search and rescue units, and humanitarian aid organizations operating in disaster-affected regions.

30 **[0106]** While particular embodiments and applications of the present disclosure have been illustrated and described in connection with survivor detection

in disaster rescue environments, it is to be understood that embodiments of the disclosure are not limited to the precise construction and compositions disclosed herein. The system 10 may be adapted for use in mining rescue operations where personnel may become trapped following tunnel collapses, in industrial accident
5 response scenarios involving structural failures, and in avalanche rescue operations where survivors may be buried under snow and debris. The radar-based detection capabilities and sensor fusion approach may also find application in security and surveillance industries, building inspection services, and infrastructure monitoring applications where detection of human presence through obstacles is desired.

10 **[0107]** Various modifications, changes, and variations can be apparent from the foregoing descriptions without departing from the spirit and scope of the disclosure as defined in the appended claims. The mobile robot chassis 100, radar sensor 110, ultrasonic sensors 120, inertial measurement unit 130, processing unit 140, and drive mechanism 150 may be manufactured using established industrial
15 processes for electronic components, sensor modules, and robotic platforms. The grid map 200 and associated confidence score algorithms may be implemented using commercially available microcontrollers and embedded processing platforms suitable for resource-constrained applications.

[0108] There are several technical advantages of the system for autonomous
20 survivor detection and navigation of the present disclosure. First, the integration of radar-based detection with probabilistic confidence mapping enables reliable survivor detection in disaster rescue scenarios where conventional visual sensors and laser-based ranging systems may fail due to dust, smoke, fog, and darkness. The millimeter-wave radar sensor is capable of detecting presence and micro-
25 motion signatures indicative of breathing patterns through obstacles such as debris and rubble, thereby extending detection capability to stationary or unconscious survivors who may not exhibit gross movement. Second, the confidence score mechanism that increases upon radar detection and decays over time in the absence of further detections reduces false positives by requiring consistent radar returns
30 before confirming a survivor location, providing improved localization accuracy compared to single-reading detection methods. Third, the sensor fusion approach

combining radar, ultrasonic, and inertial measurement data enables robust navigation and detection without reliance on simultaneous localization and mapping, camera data, or light detection and ranging data, thereby reducing computational complexity and enabling deployment on resource-constrained processing platforms. Fourth, the threshold-driven state machine architecture governing transitions between exploration, confirmation, and logging states enables structured decision-making that balances exploration efficiency with detection accuracy by requiring verification before logging survivor locations. Fifth, the grouping and ranking mechanism that identifies candidate survivor locations by clustering neighbouring high-confidence cells and ranks them based on aggregate confidence scores enables rescue teams to prioritize response efforts according to detection confidence levels. Sixth, the periodic control loop executed at a fixed rate provides consistent and predictable system behaviour with real-time responsiveness to changing environmental conditions. These technical advantages collectively contribute to an autonomous mobile robotic system for survivor detection and navigation that is both effective in low-visibility disaster environments and adaptable for deployment across a variety of search and rescue applications.

[0109] It will be appreciated that the above-detailed description, along with FIGs, is provided to illustrate the architecture and operation of exemplary embodiments of the present invention. Various modifications and enhancements can be made without departing from the scope of the invention. All such variations, as would be recognized by those skilled in the art, are intended to be within the scope of the present disclosure, which is defined by the claims that follow.

[0110] It will be appreciated by persons skilled in the art that while the invention has been described with reference to specific embodiments and accompanying drawings, numerous modifications, substitutions, variations, and equivalents are possible without departing from the spirit or scope of the present invention. The use of singular terms such as “a,” “an,” and “the” is not intended to limit the disclosed elements to a single instance unless explicitly stated, and such terms should be interpreted to include plural forms as applicable. Likewise, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” and

their grammatical variants are intended to be open-ended and non-limiting, and should be interpreted as meaning “including but not limited to.” Any enumerated listing of components, features, or elements does not imply that the items are mutually exclusive unless expressly stated.

5 **[0111]** It is further understood that where features, characteristics, or elements are described in connection with a Markush group or a disjunctive phrase (e.g., “A or B”), the invention includes all possible combinations and sub-combinations thereof unless explicitly excluded. Thus, a statement referring to “at least one of A, B, and C” should be interpreted to mean any one or more of A, B, and C,
10 individually or in any combination. The language used in this specification is selected primarily for clarity and illustrative purposes and should not be construed to limit the inventive scope unless specifically recited in the claims. Accordingly, the foregoing description of embodiments should be regarded as illustrative and not restrictive, with the scope of the invention being defined solely by the appended
15 claims and their legal equivalents

[0112] It will be recognized that various features, elements, and combinations thereof described herein may be desirably adapted for alternative implementations or other applications. Many such modifications, substitutions, enhancements, or equivalents may become apparent to those skilled in the art upon reading this
20 disclosure and are considered to fall within the scope and spirit of the present invention. The invention is not limited to the precise configurations or exemplary embodiments set forth in this specification, and various alternatives may be used without departing from the intended objectives. The claims, and not the detailed description, shall define the legal scope of protection afforded by the present
25 disclosure.

[0113] In interpreting the specification and claims, all terms should be construed in the broadest reasonable manner consistent with the context and the understanding of those skilled in the art. The use of terms such as “comprises,” “comprising,” “includes,” “including,” or “has” should be interpreted to be non-
30 exclusive and not limited to the stated elements alone. Moreover, references to “one embodiment,” “an embodiment,” or “in certain embodiments” are not meant to

imply that such features are required or exclusive, and any feature described in connection with one embodiment may be used in combination with other features or embodiments unless clearly prohibited or contextually inconsistent.

5 [0114] It should be further appreciated that reference throughout the specification to features, operations, or components using singular terms should not be construed as excluding a plurality thereof unless the context expressly indicates otherwise. Similarly, features described in the context of grouped or listed elements (e.g., A, B, and C) should be understood to encompass individual, multiple, or all combinations of said elements. Where ranges are given, all intermediate values and
10 subranges are understood to be disclosed as if specifically recited. The scope of the invention should therefore be construed as inclusive of all such combinations and logical extensions as would be appreciated by a person of ordinary skill in the relevant technical field

[0115] In another embodiment, the above disclosure is a description of the
15 invention and is not intended to limit the scope of the invention. Other variations and modifications of the above-described embodiment shall be apparent to those skilled in the art and are intended to fall within the scope of the invention as defined in the following claims.

Dated This day of May 2026

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(Kuldeep Singh)

Authorized Agent for the Applicant,
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CLAIMS

WE CLAIM:

1. A system (10) for autonomous survivor detection and navigation in a disaster rescue environment, comprising:
 - 5 a mobile robot chassis (100);
 - a radar sensor (110) disposed on the mobile robot chassis (100) and configured to detect presence and micro-motion signatures within a detection range;
 - a plurality of ultrasonic sensors (120) distributed around the mobile robot chassis (100) and configured to provide distance measurements for obstacle
 - 10 detection;
 - an inertial measurement unit (130) disposed on the mobile robot chassis (100) and configured to track heading data for dead-reckoning navigation;
 - a processing unit (140) communicatively coupled to the radar sensor (110), the plurality of ultrasonic sensors (120), and the inertial measurement unit (130),
 - 15 the processing unit (140) configured to:
 - maintain a grid map (200) comprising a plurality of cells, each cell having an associated confidence score;
 - update the confidence score of a given cell based on radar detections from the radar sensor (110), wherein the confidence score increases in
 - 20 response to a radar detection associated with the given cell and decays over time in an absence of further radar detections; and
 - control navigation of the mobile robot chassis (100) based on the confidence scores and the distance measurements from the plurality of ultrasonic sensors (120); and
 - 25 a drive mechanism (150) operatively connected to the processing unit (140) and configured to move the mobile robot chassis (100) based on navigation commands from the processing unit (140).
2. The system (10) of claim 1, wherein the processing unit (140) is further configured to operate a state machine comprising:
 - 30 an exploration state in which the mobile robot chassis (100) navigates the environment using wall-following and dead-reckoning;

a confirmation state in which the mobile robot chassis (100) performs a directional sweep to gather additional radar readings when the confidence score of a cell exceeds a first threshold; and

5 a logging state in which the processing unit (140) records a confirmed survivor location when the confidence score remains above a second threshold during the directional sweep.

3. The system (10) of claim 2, wherein the processing unit (140) is configured to transition from the exploration state to the confirmation state when the confidence score of a cell exceeds the first threshold, and to transition from the
10 confirmation state to the logging state when a confirmation criterion is satisfied.

4. The system (10) of claim 1, wherein the processing unit (140) is configured to update the confidence score of the given cell according to a formula that increases the confidence score by an increment value when a radar detection is associated with the given cell, and decreases the confidence score by a decay value
15 multiplied by an elapsed time interval when no radar detection is associated with the given cell.

5. The system (10) of claim 1, wherein the plurality of ultrasonic sensors (120) comprises a front ultrasonic sensor (120), a left ultrasonic sensor (120), and a right ultrasonic sensor (120), and wherein the processing unit (140) is
20 configured to implement a wall-following exploration strategy using the distance measurements from the plurality of ultrasonic sensors (120).

6. The system (10) of claim 1, wherein the processing unit (140) is configured to control navigation of the mobile robot chassis (100) without reliance on simultaneous localization and mapping, camera data, or light detection and
25 ranging data.

7. The system (10) of claim 1, wherein the radar sensor (110) comprises a millimeter-wave radar module configured to detect micro-motion signatures indicative of breathing patterns.

8. The system (10) of claim 1, wherein the processing unit (140) is
30 configured to identify candidate survivor locations by grouping neighbouring cells

of the grid map (200) having confidence scores that exceed a detection threshold, and to rank the candidate survivor locations based on aggregate confidence scores.

9. The system (10) of claim 1, wherein the processing unit (140) is configured to execute a periodic control loop at a fixed rate, wherein each iteration
5 of the control loop comprises receiving radar return data, updating confidence scores for cells of the grid map (200), receiving distance measurements from the plurality of ultrasonic sensors (120), and generating navigation commands.

10. The system (10) of claim 2, wherein the directional sweep comprises the radar sensor (110) sampling across a plurality of angular directions relative to a
10 current heading of the mobile robot chassis (100) as determined by the inertial measurement unit (130).

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ABSTRACT

AUTONOMOUS RADAR-GUIDED SURVIVOR DETECTION AND
NAVIGATION SYSTEM

A system (10) for autonomous survivor detection and navigation comprises a
5 mobile robot chassis (100), a radar sensor (110) configured to detect presence and
micro-motion signatures, ultrasonic sensors (120) configured to provide distance
measurements for obstacle detection, and an inertial measurement unit (130)
configured to track heading data for dead-reckoning navigation. A processing unit
(140) is coupled to the radar sensor (110), ultrasonic sensors (120), and inertial
10 measurement unit (130), and maintains a grid map (200) comprising cells each
having an associated confidence score. The processing unit (140) updates the
confidence score based on radar detections, wherein the confidence score increases
upon radar detection and decays over time absent further radar detections, and
controls navigation based on the confidence scores and distance measurements. A
15 drive mechanism (150) moves the mobile robot chassis (100) based on navigation
commands from the processing unit (140).

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